

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/339699161>

From Knowledge and Skills to Digital works: An Application of Design Thinking in the Information Technology Course

Article in *Thinking Skills and Creativity* · March 2020

DOI: 10.1016/j.tsc.2020.100646

CITATIONS

142

READS

3,623

4 authors, including:



Lin Lin

University of California, Berkeley

2 PUBLICATIONS 143 CITATIONS

[SEE PROFILE](#)



Rustam Shadiev

Zhejiang University

209 PUBLICATIONS 5,562 CITATIONS

[SEE PROFILE](#)



Wu-Yuin Hwang

National Dong-Hwa University; National Central University

247 PUBLICATIONS 5,559 CITATIONS

[SEE PROFILE](#)

From Knowledge and Skills to Digital works: An Application of Design Thinking in the Information Technology Course

Lin Lin^a, Rustam Shadiey^{a,*}, Wu-Yuin Hwang^b, Shusheng Shen^a

^a *School of Educational Science, Nanjing Normal University, Nanjing, China;*

^b *Graduate Institute of Network, Learning Technology, National Central University, Taiwan*

ABSTRACT

Design thinking, an innovative methodology, is used to guide the instruction of traditional subjects to cultivate students' 21st century skills. However, little research has clearly explained whether incorporating design thinking into the instruction of traditional subjects affects the achievement of curriculum goals. Therefore, this study aimed to explore the effects of design thinking during the instruction, in terms of reaching curriculum goals. We modified the existing design thinking framework and applied it to the information technology course. Through the experimental design and case study, we investigated how design thinking facilitated achieving the different levels of curriculum goals. Two seventh-grade classes with different instructional designs of informational technology curriculum were observed: i) one studied using the design thinking approach, and ii) another using traditional teaching methods. The results suggested that design thinking can promote the achievement of both low-level curriculum goals (i.e. mastery of the basic information knowledge and skills) and high-level curriculum goals (i.e. production of valuable, diverse, and original digital works).

Keywords: design thinking, information technology course, digital work, curriculum goal, secondary school education

* Corresponding author.

E-mail address: rustamsh@gmail.com (Rustam Shadiey) linlin.edutech@gmail.com (Lin Lin)

Introduction

Design thinking is an innovative methodology that ‘relies on a complex of skills, processes, and mindsets’ (Goldman & Kabayadondo, 2017, p. 3). It helps learners make novel digital works and develop creative thinking. Rowe (1987) described design thinking as a process and method. Buchanan (1992) explained that design thinking is a creative method that enables learners to solve wicked problems. In a famous book *The Sciences of the Artificial*, Simon (1969) emphasized the significant importance of design thinking for education. Therefore, design thinking attracted the attention of educators and researchers.

Nowadays, design thinking is employed in K-12. For example, the projects from Design for Change (2016) and Global Minimum aims to help students address real problems in their local communities through design thinking (Diefenthaler, Moorhead, Speicher, Bear, & Cerminaro, 2017). Many educators in K-12 pay much attention to developing 21st century skills in their learners. These skills are divided into three categories, i.e., life and career skills, learning and innovation skills (e.g., creativity and innovation, communication, collaboration, and problem solving), information, media, and technology skills (Trilling & Fadel, 2009). Existing studies on design thinking in K-12 focus on cultivating the 21st century skills, especially creativity, which is viewed as a core of the 21st century skills (Henriksen, Richardson, & Mehta, 2017). However, scholars often overlook the influence of design thinking on the curriculum goals achievement such as acquiring basic knowledge and skills. Besides, scholars paid little attention to the relevance of the learning process to real world experiences. Simon (1969) held the view that the primary mission of schools is to guide students to think about how to design products that are relevant to those that they experience in the real world. This kind of learning is in line with design thinking and is referred to as meaningful learning (Simon, 1969). The meaningful learning occurs ‘when students build the knowledge and cognitive processes needed for successful problem solving’ (Mayer, 2002) and it is often contrasted to rote learning. However, rote learning is a common phenomenon in K-12 due to outdated pedagogy and learning methods (Gee, 2005).

To address the above-mentioned problems, we modified existing design thinking frameworks. In addition, we devised a learning project based on the modified design thinking framework in order to apply it to the information technology (IT) classrooms in secondary school. According to the Basic Education IT Curriculum Standards of China for the year of 2012 [1], we divided the curriculum objectives into two levels: the low-level curriculum goals - they refer to mastering the basic information knowledge and skills; and the high-level curriculum goals - they refer to promoting the value, diversity, and originality of students' digital works. Arieti (1976) believed that originality is one characteristic of creativity. Originality usually refers to uncommon and unconventional products. Csikszentmihalyi (1996) mentioned that originality is rooted in reality but not fantasy, which emphasizes novelty in real life. For digital works, the value and originality are related to the high-level curriculum goals such as process and method, emotional attitude and values, i.e. learners use technology to solve real problems and improve learning/life quality with innovative solutions. The value and originality evaluate individual digital works, while the diversity evaluates group digital works. The value, diversity, and originality are related to the 'creativity', i.e. the innovation of the digital works required in the curriculum standards. We devised a learning project, in which we applied design thinking to the process of learning and instruction in order to explore whether it can achieve the curriculum goals in the IT courses of the seventh grade students. Throughout this paper, the term 'digital works' is referred to the digital works made by students when they learned the knowledge and skills of software (e.g. Microsoft Office®, WPS Office®, etc.) in the IT course.

Literature review

The Hasso Plattner Institute of Design at Stanford University (D.School) is one of the main education institutions that focuses on promoting design thinking in education. Originally, the curriculum of the D.School is designed for university students and aims to develop their problem solving and creativity skills. Recently, D.School established two laboratories (i.e. the K-12 laboratory and the Research in Education & Design laboratory) to research design thinking in K-12 educational settings. To date, there are also many other initiatives to promote and explore design thinking as a new learning model or learning paradigm in the

university or K-12 classrooms (e.g., Norman, 2000; Kwek, 2011; Wrigley & Straker, 2015; Noel & Liub, 2017). Some of these initiatives focus on the development of students' 21st century skills (Koh, Chai, Benjamin, & Hong, 2015a; Goldman & Zielezinski, 2016).

Courses for the training of design thinking

Based on the design thinking process, educators develop different specialized courses to teach the concept and skills of design thinking and cultivate innovative capability of students. For example, Grammenos & Grammenos (2017) designed a crash course named Future Designer to introduce concepts and practices of design thinking to students. His research results showed that the course appealed to different groups (i.e. teachers, children and parents). In addition, the course encouraged children to engage in creative activities and generate creative ideas. Another design thinking course was carried out in which students helped their teacher effectively use new media in the classroom (Scheer, Noweski, & Meinel, 2012; Noweski et al., 2012). The results of the study showed that the students developed various skills (e.g. metacognitive) in the course, their learning motivation was high, and their learning process was constructive. Scholars in another study designed the special curricula and found that project factors, environmental factors, and learning and teaching factors contributed to the development of the growth mindset of the students (Zupan, Cankar & Setnika Cankar, 2018). In the IT course, students' innovative capability is reflected in their digital works. To enhance the creativity of students' digital works, we devised the learning project for the IT course based on design thinking.

Taking design thinking to different subject courses

Some scholars viewed design thinking as a pedagogy (Luka, 2014; Anderson, Adam, Petray, Taylor & Otto, 2014). Educators redesigned their traditional curriculum on different subjects, e.g. literature (Carroll, 2014), mathematics (Middle-school Mathematics through Applications Project, 2009), geography (Carroll, Goldman, & Royalty, 2010), history (Koh Chai, Wong, & Hong, 2015b), etc., based on the design thinking pedagogy. For example, the Cinderella Challenge project for K-12 integrated design thinking and literature

(Carroll, 2014). This project aimed to develop students' empathic ability and was based on students' deep understanding of the fairy tale 'Cinderella'. Students role-played different characters of the fairy tale in order to understand the needs of the characters, and then used a scaffolding tool to record the needs of the characters and to provide supporting evidence. Another project (i.e. Antarctica), designed by researchers from Middle-school Mathematics through Applications Project (2009), was related to mathematics. In this project, students designed a new research station by considering all the needs of scientists that were going to live and work at the station. During the project, students acquired mathematical knowledge (e.g. concepts, scales, measurements, calculations, surface area, etc.) and then designed their stations with minimum cost. Every piece of new mathematical knowledge learned by students could help improve the design of the station, e.g. learning about how to calculate different surface areas could be useful in reducing heat loss of the station. Carroll, Goldman, & Royalty (2010) implemented design thinking in a middle school geography class. Students learned about design thinking first, e.g. they learned how to brainstorm problems and solutions to them and practiced this skill in a context. After that students used their skills to identify major problems (e.g. systems in and around the school site such as the food or traffic systems and any needs for their improvement) and solve them. Koh Chai, Wong, & Hong (2015b) developed a project that aimed to teach Singapore's history in the nineteenth century. Students were asked to design a coolies' house and they participated in the project playing different roles, such as a civil engineer, project manager, health advisor, and interior designer. The study revealed that design thinking provided many opportunities to develop students' meaningful use of ICT, creative thinking, and collaboration.

The abovementioned examples show how design thinking can be effectively embedded in a traditional curriculum. Researchers concluded that design thinking improved students' participation and set of different skills, e.g. presentation, problem-solving, cooperation, etc. Cook and Bush (2018) reviewed and synthesized the existing literature on the use of design thinking in the K-12 classroom. Based on their results, Cook and Bush (2018) argued that design thinking not only provided the teacher with a new approach to teaching content but also students with new experiences of interdisciplinary learning.

In summary, the results from the abovementioned studies demonstrate that a design thinking based curriculum is oriented on creating works by students and facilitating their real life problems solving abilities (Aflatoony, Wakkary, & Neustaedter, 2017). The abovementioned studies outlined a critical role that design thinking plays in education, e.g. developing students' 21st century skills (especially creativity and innovation), and provide useful guidelines for us to design this present study.

What else we found from the review of earlier studies on design thinking in K-12 is that it is still not clear whether design thinking can help achieve the curriculum goals as well as how to better integrate curriculum standards, learning content and design thinking (Apeddoe & Schunn, 2013). Goldman & Zielezinski (2016) connected design thinking practices with the Common Core Standards and the Next Generation Science Standards. They suggested that a critical mass of these standards was well aligned to the design thinking process and mindsets. Although they argued that teachers who use design thinking in the classroom have a good chance to achieve their curriculum goals, the data supporting such recommendations are missing. Furthermore, Carroll, Goldman, & Royalty (2010) claimed that it is still unclear how to integrate curriculum standards, lecture content and design thinking.

Research question

The K-12 curriculum of IT course needs urgent reforms. In our literature review, we did not find studies on design thinking implementation in the IT course of K-12. In China, most of IT teachers used to focus on teaching knowledge and skills in class. In addition, students create digital works without originality but with the same quality because they follow their teacher's guidance to complete the work. This kind of teaching method can aggravate students' imitation rather than innovation. Existing studies have suggested that design thinking is of great significance to pedagogy. As a result, this present research aims to address the following two questions:

- (1) Whether design thinking can enhance the learning of basic information knowledge and skills or not?
- (2) Whether design thinking can promote the value, diversity, and originality of students' digital works?

Theoretical framework

Design thinking is a well-established approach. Different scholars proposed various core processes of design thinking. Some processes fit this present study well; however, some of them were not applicable to us. Therefore, we carefully studied different design thinking processes, selected the most appropriate and relevant, and modified design thinking framework to make it applicable to this present study.

On the one hand, the differences of design thinking processes depend on occupation types, e.g., businessmen (Carroll, 2013), administrators (Bell, 2008), teachers (Radford University, 2013; IDEO, 2013), and students (D.School, 2018; Goldman, Roth, & Carroll, 2009). As shown in Table 1, although there are different, specific processes of design thinking, the main processes can be generalized into the following three, i.e., inspiration, ideation, and implementation (Brown, 2008). Brown (2008) stated that inspiration happens in circumstances that motivate the search for solutions; ideation means generating, developing and testing ideas; implementation refers to chart a path to marker. Although Brown (2008) introduced these elements from a business perspective, it has also been applied to different fields, such as education (Grift & Kroeze, 2016). Different design thinking processes are a kind of ‘localization’ of these three elements in a specific occupation.

On the other hand, the non-linearity of elements and the iteration of the design process are two typical characteristics of design thinking. However, most of the existing design thinking processes are linear visually, they also emphasize the non-linearity and iteration through drawing lines and arrows in the process. Without careful observation of those processes, it is difficult to capture these important characteristics in the existing design thinking process.

Table 1. The core elements of the design thinking process (synthesised from Carroll, 2013; Bell, 2008; Radford University, 2013; IDEO, 2013; D.School & Thomas, 2017; Goldman, Roth, & Carroll, 2009).

Items	Inspiration	Ideation	Implementation	
SAS Institute	Scope, Explore	Ideate, Synthesize	Roadmap	Value

Bell	Understand, Observe	Visualize, Evaluate/Refine	Implement	Evaluate/Refine
Radford University	Discover and Define, Generate Empathy	Imagine-Explore-Create	Implement and Evolve	
IDEO	Discovery, Interpretation	Ideation	Experimentation	Evolution
D.School	Empathize, Define	Ideate	Prototype	Test
D.School	Understanding, Observation, Define	Ideate	Prototype	Test

We studied different design thinking processes and collected the most appropriate to solve issues of existing design thinking processes and to fit IT education (Figure 1). Our framework visualizes non-linearity and iteration through an iteration in an iterative way consisting of three phases. Each phase is also based on the above three main elements (i.e., in design thinking processes, analysing users' needs would not only fall under inspiration but also ideation), which reflects the non-linearity of the proposed framework. Besides, we separated evaluation from implementation as an intermediate phase to facilitate iterations.

Inspiration: real-world problem

The first phase of the design thinking framework focuses on real-world problem definition. During this phase, the main task includes three parts: understanding the project theme (inspiration), analysing user's needs (ideation), and determining a theme of their digital work (implementation).

Ideation: design scheme

The ideation phase aims to generate a design scheme. Students first investigate the user's needs in-depth and find a digital work that best fits this need (inspiration). Then, they analyse the basic components of the digital work and collect corresponding materials (ideation), followed by the design scheme determination of the digital work (implementation).

Implementation: digital work

This phase is associated with creating digital works. Students explore the user's needs again and make sure the above design scheme meets the user's needs (inspiration). If there are still some problems in the design scheme, students need to solve them by improving the design scheme (ideation). In the end, they create the final digital works (implementation).

Evaluation: process and outcome

Inspired by the research of Lindberg, Meinel, and Wagner (2010), we separate the design thinking process into two following spaces: the problem space and the solution space. In addition, our design thinking process includes the evaluation phase, because it can facilitate the iterative alignment of both spaces. The two spaces interact and affect each other through the evaluation. Specifically, the evaluation phase aims to improve the originality of elements, materials, and compositions of digital works.

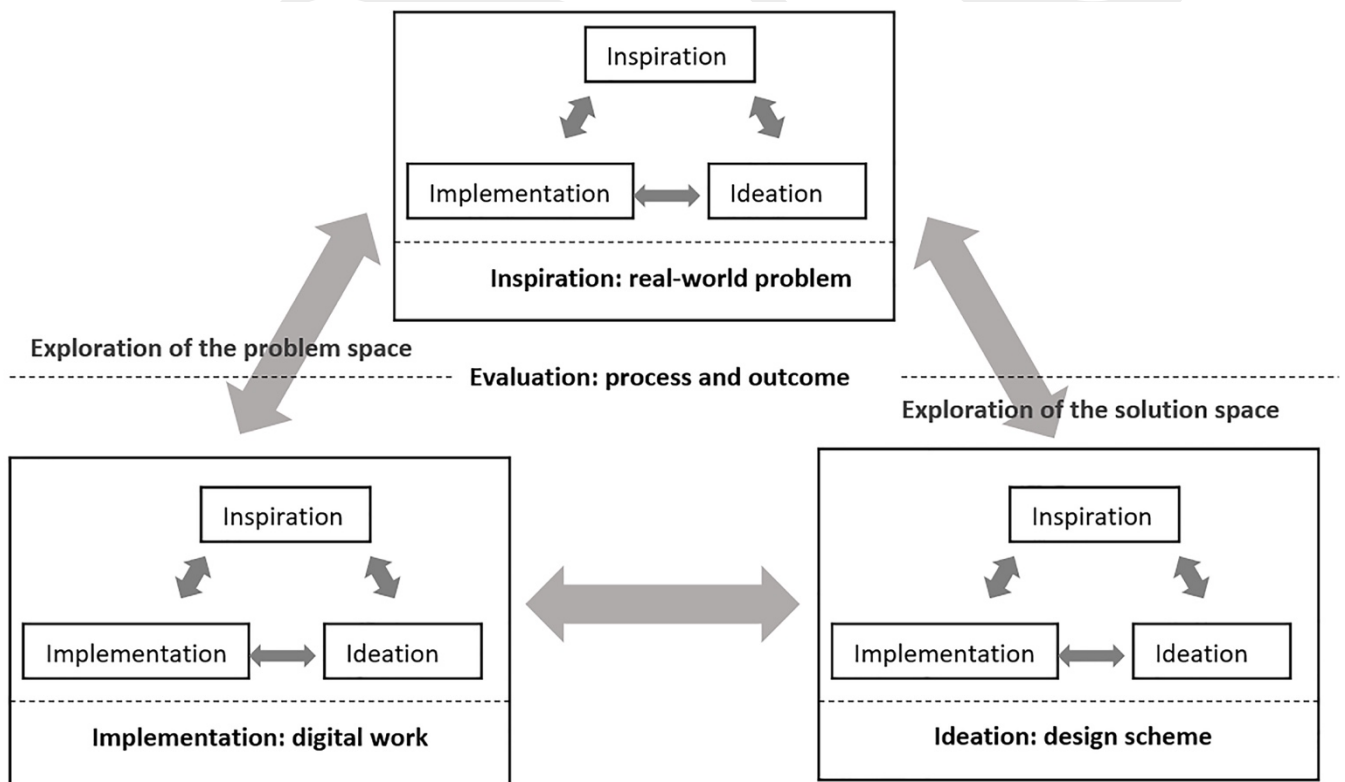


Figure 1. The design thinking framework for IT course

Method

The researchers collaborated with four experienced instructional designers and educational technologists. Three seminars (each seminar lasted one an hour) were organized to design this research and to ensure the validity and reliability of our research instruments and collected data. Furthermore, a pilot study, i.e. a small scale preliminary study, was conducted in order to evaluate the feasibility of the key steps in a future, full-scale study. The IT course was administered and twenty-six fourth grade students in Chinese public school participated in this pilot study. They were asked to design a container based on their daily life experience using 3D printing technologies by following the design thinking framework. Lesson plans and students' digital works were analysed. The results were positive suggesting that learning performance based on the design thinking approach promoted the diversity of students' digital works and helped them come up with design ideas according to the user's needs.

Sites and participants

Full scale study followed a pilot study. It was also conducted in China, at a public school. The participants were seventh-grade students aged between 12-14 years old. The researcher selected two classes for this study; therefore, we followed a convenience sampling design rather than a random sampling one. There were 33 students in class A (20 boys and 14 girls) and 29 students in class B (15 boys and 14 girls). Students in each class were divided into six groups (group A-F). Based on the interview data with their instructors, the participants in both classes had very low prior knowledge on IT and design thinking because very few students have ever learned any IT subject before this study and this was their first experience with the design thinking approach.

Two teachers participated in this study. They were of similar age, the same sex, and the same education level (Master of Science). One teacher instructed students in class A and the other in class B. The teacher from class A has been studying design thinking for two years.

Class A students learned using the design thinking approach and class B students learned using traditional teaching methods. Both classes took seven lessons (each lesson lasted 40 minutes) in which they learned the same unit. The knowledge and skills in Class B were arranged in the order of the textbook whereas those in Class A were arranged according to the project. Details of the learning activities for Class A are provided in the Project description section. As for Class B, each lesson in this unit was taught independently. For example, the Beautify Document was one lesson of Class B and the goal of this lesson was to master the skills of adding an image, word art, and auto shape in WPS Writer®. After the teacher demonstrated how to add an image or shape to a document, students designed their digital works (e.g. a picture and an introduction of Santa Claus).

Project description

Under the design thinking framework (Figure 1), a project named ‘New Year’s Party Preparation’ was designed by the IT teacher and researchers. It aimed to encourage students to solve a real-world problem about the preparation of a party by using IT. Specifically, we selected a unit, named Application Documentation Design and Creation, from the IT textbook and students were asked to design digital documents for this party (e.g. to make posters for party promotion).

The following were the low-level curriculum goals of this unit: (1) to learn how to change the default font and paragraph spacing in WPS Writer®; (2) to learn how to convert the format of the page in WPS Writer®; (3) to learn how to add an image, word art, and text box in WPS Writer®; (4) to learn how to add a table and put text in a table in WPS Writer®; (5) to learn how to create a mind map and insert map elements in Mind Mapping Software; and (6) to learn how to add markers with Mind Mapping Software. The goals from one to four were related to using WPS Writer®, while the goals five and six were related to using Mind Mapping Software.

The high-level curriculum goals of this unit included the following: (7) to find the needs of different students in preparing or attending a party through empathy; (8) to use WPS Writer® and Mind Map

Software for creating a product that meets those needs through brainstorming and other collaborative activities based on design thinking; and (9) to develop students' awareness of improving learning and life by using WPS Writer® and Mind Map Software. Note that, in this project, students learned IT concepts and skills of design thinking by experiencing the process of design thinking rather than through formal design thinking course.

The designed for this study project consisted of three stages and seven tasks. At the end of each task, students were required to evaluate and improve their digital works. The first stage contained four tasks: (Task 1) to analyse the compositions, disadvantages of an existing document named Activity Plan for the New Year Party (e.g. the layout of this document is bad) and to understand the project's background; (Task 2) to think about the versatile applications of documents in New Year's party preparations and make a mind map in group. The above two tasks match goals 1, 5, and 7. In this stage, students became more familiar with the background and knew the various possibilities of IT to solve real problems.

The next task, (Task 3) to match goals 2, 3, and 7 students read profiles and made their own profiles. This task aimed to develop students' empathy through several activities. Firstly, three profiles designed in advance were provided to the student to explore the needs of different people and to analyse the compositions of each profile, which are also indispensable in their own profiles. Each profile included a photo and some information, for example, one of the profiles included the following message: 'Hello everyone! I am Shen Yi, 15 years old, from Grade eight. I am the president in my class and responsible for organizing the party. Volunteers complained to me that they didn't know what their job was. How can I help them?' Secondly, students interviewed their teachers, classmates or other people who were going to the party to collect information about their needs. In the previous example, Shen Yi's need was to help volunteers know their job. Finally, there were some questions for students to inspire their ideas, such as 'if you want to invite a friend to the party, what information would you tell him or her?' After undergoing the above activities, students proposed their needs as a guest or organizer and made their profiles. (Task 4) To match goals 4 and 7 students summarized each group member's need/s supported by a scaffolding tool, 'user

+ need + insight’, recorded it in a table, and selected one as the theme of their final digital work. Here is an example of the scaffolding tool use: ‘Shen Li (user) needed a task schedule to help volunteers know more information about their job (need) because I knew from his profiles that he wanted to resolve volunteers’ complaints (insight).’

The next stage contained two tasks: (Task 5) according to the theme of the digital work, each group first explored the constituent elements of their digital works by brainstorming and presented these elements in the mind map. This task matched goals 6 and 8 and aimed to find innovative elements for better preparation of the digital work. (Task 6) To match goals 4 and 8, students made a tabular checklist that contained all the collected digital works’ constituent elements. According to the checklist, students collected creative materials, specified the digital work design scheme and conceived the layout of the digital work.

In the final stage, (Task 7) to match all the curriculum goals of this unit and to develop students’ comprehensive skills and iterative optimization, students made the final digital work according to their design scheme in an iterative fashion. The iteration means that students presented their design first in the group, then in the class, to receive some feedback, and then improve it.

Data collection and analysis

The general scale to understand if students did well in this unit was divided into two parts. Both parts were related to two research questions respectively. One was the mastery of basic information knowledge and skills, the other was the value, diversity, and originality of students’ digital works.

To address the first research question, we evaluated students’ digital works, compared their learning performance with prior knowledge to evaluate how well students grasped knowledge and skills, tested the difference between two groups in terms of their scores of digital works (Table 2). Creswell (2017) suggested that the independent sample t-test can be used to test for a difference between two groups in terms of one dependent variable. We employed the independent sample t-test to determine the statistical difference between two classes on their perception of basic information knowledge and skills. To ensure the objectivity

of scores, we evaluated mastery of basic information knowledge and skills of the students' digital work based on rubrics presented in the Knowledge and skills column of Table 2. Possible scores for each rubric are presented in the Possible score column. For example, for "New Year Party Plan" (Table 2) task, if students changed both font and paragraph, their score was 100; if they only changed the font or the paragraph, the score was 50; and if students did not change anything, their score was 0. Teachers were allowed to add or subtract 16 or 25 points from the rubric according to the digital works. Two teachers scored students' digital works and the reliability coefficient of the score was significant (Table 2). We selected digital works that involved similar knowledge and skills from classes A and class B for comparison. There were three pairs of data that were comparable in this study: pairs I (No. 1 and No. 8), pairs II (No. 3 and No. 9), and pairs III (No. 5 and No. 10).

To answer the second research question, we took a case study. This approach was viewed as a form of qualitative research and used to describe and interpret data by collecting and analysing multiple sources of information (Denzin & Lincoln, 2000). We compared the digital works (Table 2) and lesson plans of the whole unit from different classes in the following three aspects: (1) value – it refers to the significance of digital works to the real world, which lies in that these digital works were able to solve the practical problems in our daily life; (2) diversity – it refers to the ampleness of various kinds of digital works' themes, as well as the diversity of expression forms and contents of the digital works within the same theme; (3) originality – it means that there were some new ideas of students expressed their digital works. Also, we could find the reasons accounting for the differences in the digital works of two classes by comparing their lesson plans.

Table 2. Students' digital works and their scores

No.	Digital work's name	Class	Document type	Knowledge and skills	Possible scores (in %)	Pairs	Average (in %)	Reliability*
1	New Year Party	A	Document	Change the default	0, 50,	I	57.58	.84

	Plan			font and paragraph spacing	100				
2	Application of Documents in Party Preparations	A	Mind mapping	create a mind map and insert map elements	0, 50, 100	-	75.76	1.00	
3	Profile	A	Document	Convert the format of the page; Add an image, word art, text box	0, 33, 66, 100	II	79.79	.72	
4	Summary of Each Group Member's Need	A	Document	Put text in a table	0, 50, 100	-	96.21	1.00	
5	Compositions of the Digital Work	A	Mind mapping	Insert map elements and add markers	0, 50, 100	III	87.88	1.00	
6	A Checklist about the Digital Work Compositions	A	Document	Add a table and put text in a table	0, 50, 100	-	77.27	.76	
7	The Final Digital Work	A	Document	All the above knowledge and skills	0, 33, 66, 100	-	81.82	.84	
8	Changing the Font	B	Document	Change the default font and other items	0, 33, 66, 100	I	51.90	.91	
9	Beautify Document	B	Document	Add image, word art, auto shape	0, 33, 66, 100	II	57.48	.92	
10	Create a Mind Mapping	B	Mind mapping	Insert map elements and add image	0, 33, 66, 100	III	65.59	.92	

*The Bivariate Correlations was used to check the reliability of students' scores.

Results

Facilitating the achievement of the low-level curriculum goals with design thinking approach

At the beginning of this study, the participants in both classes had very low prior IT knowledge. The results of students' digital works evaluation are shown in Table 2 (see Average column). According to the results, the knowledge and skills of the participants in both classes were promoted. The results of the comparison between two classes with respect to their digital works scores are presented in Table 3. There was not a significant difference in scores between two classes for Pair I ($p>0.05$). Non-significant differences between the groups for Pair I could be ascribed to the fact that it was the first learning activity in this present study, carried out at the very beginning of the study, and the scores used for the comparison were collected not long after the study began. Thus, it is difficult to see the influence of design thinking approach in the initial stage of implementation. The difference in scores between the two classes was statistically significant for Pairs II and III ($p<0.05$). That is, the scores of Class A (Mean=79.79) were significantly higher than those of Class B (Mean=57.48) in Pair II. In addition, the scores of Class A (Mean=87.88) were higher than those of Class B (Mean=65.59) in Pair III. These results may suggest that the knowledge and skills of the participants were promoted because of the design thinking approach.

Table 3. Comparison of the differences in scores between two classes

Pair	Class	Mean	Standard deviation	T-test
I	A	57.58	32.766	.637
	B	51.90	37.431	
II	A	79.79	39.925	1.935*
	B	57.48	49.521	
III	A	87.88	18.876	2.783**
	B	65.59	39.345	

* $p<0.05$ ** $p<0.01$

Facilitating the achievement of the high-level curriculum goals with design thinking approach

As we mentioned, the participants in both classes had very low prior IT knowledge. The result of students' digital works evaluation is reported in Table 4. The results in the tables show that the IT knowledge and skills of the participants were promoted. However, the difference between the two classes was apparent. After comparing digital works and lesson plans from Class A and Class B, we found differences in the value, diversity, and originality of digital works. That is, value, diversity, and originality of digital works of Class A were better. The results may suggest that design thinking approach was beneficial in facilitating the achievement of high-level curriculum goals.

Table 4. Comparison of the differences in digital works and lesson plans between two classes

Pair	Class A	Class B
Value	The themes of digital works are determined by the students and inspired by real requirements.	The teacher according to knowledge and skills determines the theme of digital works.
Diversity	Different groups have different themes and different digital works designed by group members.	The theme and form of students' digital works are the same.
Originality	Students' original digital works.	Digital works made by students but design by the teacher.

Value: the themes of digital works were inspired by real requirements

In Class B, themes of students' digital works were determined by the teacher according to knowledge and skills. For example, one of the themes was the Introduction of Santa Claus (this was introduced earlier by us in the Sites and participants section). The teacher in Class B aimed to help students learn the knowledge and skills of IT but not develop their problem solving skills. Unlike Class B, themes of digital works identified by six groups in Class A included Party Notes, Party Schedule, Worklists of Party Preparation, Propaganda Poster of the Party, and Program List of the Party. From students' digital works named Summary of Each Group Member's Need, it is possible to find that the themes which students wanted to select were all inspired by practical needs in the party preparation. For example, there was the following message in the

profile created by one student: ‘Hi everyone, I am Chang Tian from grade seven. I also a volunteer and plan to attend the preparation of the New Year party and would like to know the matters that need our attention during the party organization.’ Group A determined its theme as Party Notes according to his needs. Another student who was an organizer of the party posted the following message on her profile: ‘Hello everyone! I am Weijin Zhu from grade seven. My school will host a New Year Party. I feel very excited about it. Many students want to know the program list of this party. Therefore, I want to get more details about the party program.’ Group D determined its theme as Propaganda poster of the Party according to her needs.

Defining the need was more important than finding the solution in design thinking. It is noteworthy to know why we describe students’ requirements as real or practical needs, and how students came up with these needs. On the one hand, students themselves were guests or organizers of the party, so that students’ requirements for this party were meaningful. On the other hand, all students presented their needs as a guest or organizer of the party and wrote these needs in their profiles. A scaffolding tool was available for the students to explore the needs of other classmates. Then they recorded all these needs and digital works in the table of Summary of Each Group Member’s Need.

Diversity: Ideation facilitates various types and forms of digital works

The themes, material, and content of student’s digital works (Figure 2) from Class B were the same and digital works were created in the same way because students were limited by the detailed steps provided by the teacher. Also, the teacher prepared the same learning material and content for all students beforehand.

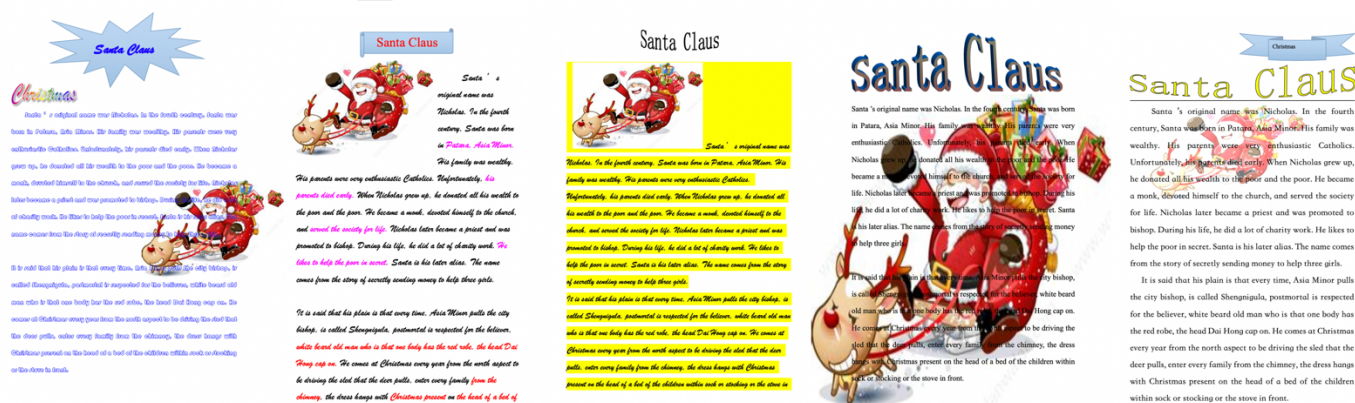


Figure 2. Some examples of students' digital works from different groups in Class B (translated from Chinese)

The situation was different in Class A. Ideation (i.e. the second stage of the design thinking framework) abled students to create digital works with diverse themes, material, and content. The main reason for this result was cooperation among students within their group and class. Ideation encouraged such cooperation within the class. At the beginning of the project, the students of each group thought about the applications of documents in the process of preparing the New Year party. The ideas may be limited and making posters was the most common answer. Inspired by the teacher, students put forward more ideas through brainstorming. For example, some documents should be presented with pictures and texts, including postcards, New Year cards, invitations, party programs, theme posters, party mascots, small notices; some other documents should be presented with tables and texts, including cast of characters, list of staff tasks, schedules, New Year's calendars. Figure 3 presents mind mapping created by six groups. The effect of ideation is obvious in the figure. That is, it provided a new and more complete perspective for students to find the themes of digital works and also enabled a diversity of themes. Besides, after finding different themes, each group could choose a theme of digital works according to their interests.

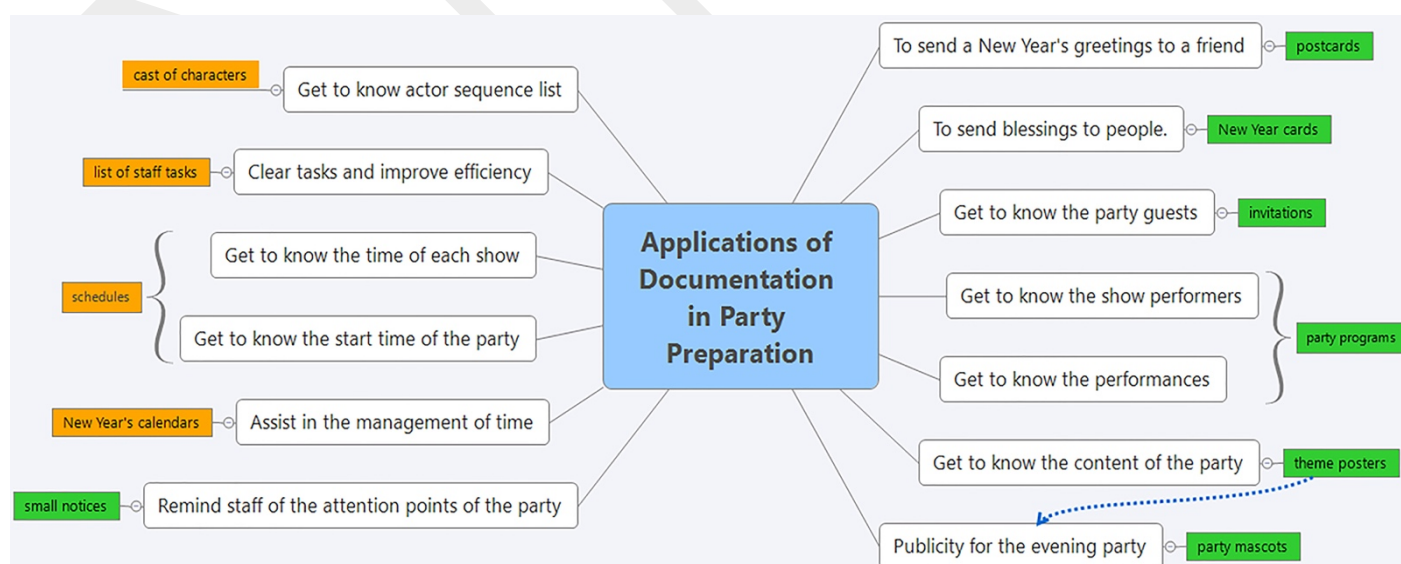


Figure 3. Applications of Documentation in Party Preparation

Teamwork was also stressed at this stage. To improve the effectiveness of group learning, during the whole project process, many tasks were elaborately designed in a collaborative way, and students were

asked to engage in the group discussion and complete those tasks together. These collaborative tasks contained requirements analysis, themes selection, determination of elements of digital works, etc. For example, the task of determination of elements of digital works can be accomplished by a single person not so good as it could be done by a group of students. There would be limited composition elements that a single person came up with. But if we collect ideas from a group then there is no doubt that the integrity of the digital work's constituent elements will be improved. For example, in the digital work of group C, the theme of work was a worklist for preparing the party. The fourth task was to determine the composition of digital works by mind mapping. At first, the students only thought about people and their job involved in the worklist. Through the brainstorming of all team members, students improved constituent elements including title, content (people and their job), background, decoration style and other aspects of the worklist.

Originality: Design thinking facilitate the originality of digital works

As mentioned earlier, students from Class B did not need to think about how to design their digital works, which led to the absence of new ideas in their digital works. As we expected, we could find novel ideas in digital works from Class A. For example, as shown in Figure 4, group B made their schedules in the form of a timeline, or represented time directly by a clock. The creativity of worklists of group C was reflected in the added cartoon figures of different people (in fact, what they wanted to use here was a real photo, but due to some restrictions, they could not collect everyone's photo) so that people could quickly find their tasks.



Figure 4. Some examples of students' final digital works from different groups in Class A: on top are schedules and on the bottom are worklists (translated from Chinese)

Conclusions and pedagogical implications

This study adopted the design thinking framework for the IT learning project. Previous studies have shown that design thinking can improve students' 21st century skills, especially creativity. From the study's results, we can see that achieving the low-level and high-level curriculum goals were facilitated by the design thinking approach. This was because the design thinking approach enabled students in Class A to be involved in different activities, including research, group cooperation, works show, etc. The IT courses based on design thinking substantially cultivate students' communication and collaboration skills, problem-solving skills, information technology skills and other skills in activities. More importantly, the research results not only showed that design thinking was conducive to the promotion of students' knowledge and skills but also showed that the design thinking process can promote value, diversity, and originality of digital works in the IT course. To some extent, this answers the question of existing research (Apeddoo & Schunn,

2013; Carroll, Goldman, & Royalty, 2010) on whether design thinking in class can achieve the curriculum goals.

The influence of design thinking on the implementation of curriculum goals cannot be separated from the design thinking pedagogies (Goldman & Zielezinski, 2016). In the research, project design based on design thinking abides by the following points.

Firstly, the topic selection of the project is very important, which determines whether students' digital works are based on realistic demands. People should be able to solve problems in the real world through digital works, which can activate students' learning motivation. Students reported that the realistic problems under this topic also motivated them to learn. Dorst (2011) claim the core of design thinking is that designers not only need to find problem solutions (how), they also need to define problems (what). 'What' and 'how' lead to not only results but value as well.

Secondly, scaffolding tools should be designed to support the design thinking process, e.g. students can use these tools for sharing their ideas and digital works. This includes not only the design thinking framework for IT courses presented in this paper but also the framework that facilitated each task, for example, scaffolding tool based on 'user + need + insight'.

Thirdly, integrating the subject knowledge into projects is very important. One project may contain a sequence of tasks, which are designed based on the design thinking framework. Each task aims to promote the production of the final digital work. Besides, each task contains a specific requirement and students need to learn new subject knowledge to address this need. With such a design, students can experience the general process of design thinking and can connect the subject knowledge in the textbook with the real world experiences through the projects. There may not be a direct relationship between subject knowledge and design thinking. However, the design thinking process improves student motivation and enables innovation as a learning process (Beckman & Barry, 2007), the mindsets of design thinking (Goldman et al., 2012) have taken on board ideas more easily, which indirectly promoted the learning of subject knowledge. As a result,

the responsibility of the teacher is no longer to teach knowledge but to design a project and guide the implementation of a project. Teachers need to design different activities to promote cooperation between students during the projects and teach students the ways to acquire knowledge instead of directly providing knowledge to students.

Some limitations need to be acknowledged and further work needs to be conducted to address them. Firstly, qualitative research requests multiple sources of information but we only collected digital works and lesson plans. In the future study, we can collect more information, such as interviews with students and teachers, class observation, and questionnaires. Secondly, the small sample size of this study prevents us from generalizing the findings to a broader population. Thirdly, since the course was related to IT, the findings are suitable for a particular subject. Further research is required to assess the implementation of the design thinking framework in various subjects. Fourthly, there may be some other factors that can be responsible for the differences in average scores found between two classes. It is popular in China that students take additional classes after school to prepare for the junior high school entrance examination. Therefore, two possible factors, need to be considered in future studies, are the amount of time the students spent on the learning activity and the amount of learning content the students covered outside of school and we could not control either of them.

Note

1. In the Basic Education IT Curriculum Standards of China for the year of 2012, the curriculum' goals are divided into three dimensions: (1) Knowledge and Skills: mainly refers to master the basic operations and applications of IT; (2) Process and Method: mainly refers to use of IT to solve real-world problems and come up with creative problem solutions; (3) Emotional Attitude and Values: mainly refers to development of the awareness of using IT to promote learning and improve life.

Acknowledgements

This work was supported by the China's National Office for Education Sciences Planning [grant numbers BCA190081]; and the Postgraduation Research and Practice Innovation Program of Jiangsu Province [grant numbers KYCX19_0728].

Author Statement

Lin Lin: Methodology, Formal analysis, Investigation, Data Curation, Writing - Original Draft

Rustam Shadiev: Methodology, Writing - Review & Editing, Visualization

Wu-Yuin Hwang: Methodology, Writing - Review & Editing

Shusheng Shen: Methodology, Conceptualization, Supervision

References

- Aflatoony, L., Wakkary, R., & Neustaedter, C. (2017). Becoming a Design Thinker: Assessing the Learning Process of Students in a Secondary Level Design Thinking Course. *International Journal of Art & Design Education*, 37(3), 438–453. doi: 10.1111/jade.12139
- Anderson, N., Adam, R., Petray, T., Taylor, P., & Otto, T. (2014). *Design thinking frameworks as transformative cross-disciplinary pedagogy: Final Report*. Sydney: Australian Government & Office for Learning and Teaching. doi: 10.13140/2.1.1068.8806
- Apeddoe, X. S., & Schunn, C. D. (2013). Strategies for success: Uncovering what makes students successful in design and learning, *Instructional Science*, 41(4), 773–791. doi:10.1007/s11251-012-9251-4
- Arieti, S. (1976). *Creativity. The Magic Synthesis*. New York: Basic Books, (Arieti collects characteristics from the work of a number of researchers in chapter 15).
- Beckman, S. L., & Barry, M. (2007). Innovation as a learning process: Embedding design thinking. *California Management Review*, 50(1), 25-56. doi:10.2307/41166415
- Bell, S. J. (2008). Design thinking, *American Libraries*, Jan/Feb, 44-49.
<http://scholarshare.temple.edu/xmlui/handle/20.500.12441/882>
- Brown, T. (2008). Design thinking. *Harvard Business Review*, 86(6), 84-92, 141.
<http://www.ncbi.nlm.nih.gov/pubmed/18605031>

- Buchanan, R. (1992). Wicked Problems in Design Thinking. *Design Issues*, 8(2), 5. doi:10.2307/1511637
- Carroll, C. (2013). *2013 Design as Strategy Forum James Malizis SAP*. Retrieved from <https://slideplayer.com/slide/10948454/>
- Carroll, M. (2014). *The Cinderella Challenge*. Retrieved from https://dschool-old.stanford.edu/groups/k12/wiki/0c2db/The_Cinderella_Challenge.html
- Carroll, M., Goldman, S., & Royalty, A. (2010). Destination, imagination & the fires within: design thinking in a middle school classroom. *International Journal of Art & Design Education*, 29(1), 37-53. doi:10.1111/j.1476-8070.2010.01632.x
- Cook, K. L., & Bush, S. B. (2018). Design thinking in integrated STEAM learning: Surveying the landscape and exploring exemplars in elementary grades. *School Science and Mathematics*, 118(3–4), 93–103. doi:10.1111/ssm.12268
- Creswell, J. W. (2017). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. SAGE Publications.
- Csikszentmihalyi, M. (1996). *Creativity. Flow and the Psychology of Discovery and Invention*. New York: Harper Collins Publishers, Inc.
- D.School, 2018. Bootcamp bootleg. Derived from <https://dschool.stanford.edu/resources/the-bootcamp-bootleg>.
- Denzin, N., & Lincoln, Y. (2000). *Handbook of qualitative research (2nd ed.)*. Thousand Oaks, CA: Sage.
- Design for Change. (2016). *Impact*. Retrieved from <http://www.dfeworld.com/research.html>
- Diefenthaler, A., Moorhead, L., Speicher S., Bear, C., & Cerminaro, D. (2017). *Thinking & Acting Like a Designer: How design thinking supports innovation in K-12 education*. Retrieved from <https://www.wise-qatar.org/2017-wise-research-design-thinking/>
- Dorst, K. (2011). The core of 'design thinking' and its application. *Design Studies*, 32(6), 521-532. doi:10.1016/j.destud.2011.07.006
- Gee, J. P. (2005). What would a state of the art instructional video game look like, *Innovate: Journal of Online Education*, 1(6), 6. <https://www.learntechlib.org/p/107271/>.
- Goldman, S. Roth B., & Carroll, M. (2009). *Taking Design Thinking to School*. Retrieved from <http://web.stanford.edu/dept/SUSE/taking-design/presentations/Taking-design-to-school.pdf>
- Goldman, S., & Kabayadondo, Z. (2017). *Taking Design Thinking to School: How the Technology of Design Can Transform Teachers, Learners, and Classrooms*. London: Taylor & Francis.

- Goldman, S., & Zielezinski M.B. (2016). Teaching with Design Thinking: Developing New Vision and Approaches to Twenty-First Century Learning. In Annetta L. A., & Minogue J. (Eds.) *Connecting Science and Engineering Education Practices in Meaningful Ways* (pp. 237-262). Cham: Springer.
- Goldman, S., Carroll, M.P., Kabayadondo, Z., et al. (2012) Assessing d.learning: Capturing the Journey of Becoming a Design Thinker. In: Plattner H., Meinel C., Leifer L. (eds) *Design Thinking Research. Measuring Performance in Context* (pp. 13-33). Springer, Berlin: Heidelberg.
- Grammenos, D., & Antona, M. (2017). Future designers: Introducing creativity, design thinking & design to children, *International Journal of Child-Computer Interaction*, 16, 16-24.
doi:10.1016/j.ijcci.2017.10.002
- Grift, T. C. V. D. & Kroeze, R. (2016). Design thinking as a tool for interdisciplinary education in health care. *Academic Medicine*, 91(9), 1234-1238.
- Henriksen, D., Richardson, C., & Mehta, R. (2017). Design thinking: A creative approach to educational problems of practice. *Thinking Skills and Creativity*, 26, 140-153. doi:10.1016/j.tsc.2017.10.001
- IDEO. (2013). Design Thinking for Educators. Retrieved from
<http://www.designthinkingforeducators.com/toolkit/>
- Koh, J. H. L., Chai, C. S., Wong, B., & Hong, H. Y. (2015a). Technological pedagogical content knowledge (TPACK) and design thinking: a framework to support ICT lesson design for 21st century learning, *The Asia-Pacific Education Researcher*, 24(3), 535–543. doi:10.1007/s40299-015-0237-2
- Koh, J. H. L., Chai, C. S., Wong, B., & Hong, H. Y. (2015b). Design Thinking for Education: Conceptions and Applications in Teaching and Learning. Singapore: Springer.
- Kwek, D. (2011). Innovation in the Classroom: Design Thinking for 21st Century Learning. Stanford University. Unpublished Master's Thesis.
- Lindberg, T., Meinel, C., & Wagner, R. (2011). Design Thinking: A Fruitful Concept for IT Development?, in Meinel, C., Leifer, L., Plattner, H. (Eds.) *Design Thinking. Understanding Innovation* (pp. 3-18). Berlin: Springer.
- Luka, I. (2014). Design Thinking in Pedagogy, *Journal of Education Culture and Society*, 2, 63-74. doi: 10.15503/jecs20142.63.74
- Mayer, R. E. (2002) Rote Versus Meaningful Learning, *Theory Into Practice*, 41:4, 226-232, doi: 10.1207/s15430421tip4104_4
- Middle-school Mathematics through Applications Project (2009). *The Antarctica project: a middle-school mathematics unit*. Retrieved from <http://web.stanford.edu/group/redlab/cgi-bin/materials/Antarctica%20Project-Design%20Thinking%20version.pdf>

- Noel, L. A., & Liub, T. L. (2017). Using design thinking to create a new education paradigm for elementary level children for higher student engagement and success, *Design & Technology Education*, 2(1), 12.
- Norman, J. (2000). Design as a framework for innovative thinking and learning: how can design thinking reform education? IDATER 2000 Conference, Loughborough: Loughborough University.
- Noweski, C., Scheer, A., Büttner, N., Thienen, J. V., Erdmann, J., & Meinel, C. (2012). Towards a Paradigm Shift in Education Practice: Developing 21st century skills with Design Thinking. In Plattner H., Meinel C., Leifer, L. (Eds.) *Design Thinking Research. Understanding Innovation* (pp. 71-94). Berlin: Springer.
- Radford University. (2013). *Design Thinking For Educators-A Radford University Methodology Workbook*. Retrieved from <https://www.slideshare.net/ccvidadmin/design-thinking-for-ed-wbk1c-for-screen-26174639>
- Rowe, P. (1987). *Design thinking*. Cambridge, MA: MIT Press.
- Scheer, A., Noweski, C., & Meinel, C. (2012). Transforming constructivist learning into action: Design Design thinking in education, *Design & Technology Education*, 17(3), 8-19. Retrieved from <https://ojs.lboro.ac.uk/DATE/article/view/1758>
- Simon, H. A. (1969). *The sciences of the artificial*. M.I.T. Press.
- Trilling, B., & Fadel, C. (2009). *21st Century Skills: Learning for Life in Our Times*. San Francisco, CA: John Wiley & Sons.
- Wrigley, C., & Straker, K. (2015). Design thinking pedagogy: the educational design ladder, *Innovations in Education & Teaching International*, 54(4), 374-385. doi:10.1080/14703297.2015.1108214
- Zupan, B., Cankar, F., & Setnikar Cankar, S. (2018). The development of an entrepreneurial mindset in primary education. *European Journal of Education*, 53(3), 427–439. doi: 10.1111/ejed.12293